

KRISHNA Koushik THOKALA

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Summary

AWS Certified Solutions Architect – Associate and recent MS CS graduate with 2 years of carrier-grade network operations experience, including leading 5-person NOC shifts supporting 25+ Tier-1 clients. Built and validated AWS and Kubernetes infrastructure using Terraform, GitOps, security controls, observability, and failure-recovery testing.

Technical Skills

Cloud Infrastructure: AWS, VPC, IAM, EKS, ECR, RDS, Lambda, DynamoDB, ALB, Terraform
Kubernetes & Delivery: Kubernetes, Docker, Helm, Argo CD, GitOps, GitHub Actions, CI/CD
Observability & Reliability: CloudWatch, Prometheus, Grafana, HPA, k6, incident response, RCA, SLOs, MTTR
Programming & Systems: Python, Bash, SQL, PostgreSQL, FastAPI, REST APIs, Linux
Networking: TCP/IP, BGP, MPLS, DNS, Security Groups, NetworkPolicies

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Network Operations Center – Shift Lead

Pune, India

- Led 5-person shifts in a 24/7 NOC supporting 25–30 Tier-1 carrier clients and a daily queue of 40+ priority incidents across BGP/MPLS, DWDM, and transport networks.
- Diagnosed high-severity BGP/MPLS, TCP/IP, and DWDM failures across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, loop testing, and segment-wise isolation; coordinated TAC, vendor, and field teams through restoration.
- Coordinated incidents from triage and escalation through customer updates, structured handoffs, restoration, and RCA documentation, supporting 99.9% availability and 4-hour restoration SLA targets.
- Earned a Certificate of Excellence for ownership and problem-solving in isolating and resolving escalated customer incidents, converting an escalation into customer appreciation.

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

GitHub

- Engineered an EKS GitOps delivery platform where 4 Argo CD Applications reconciled Git-managed Helm releases across isolated dev, staging, prod, and observability namespaces, including Argo CD-managed Prometheus and Grafana.
- Provisioned VPC, EKS, ECR, IAM, and AWS Budgets with Terraform; secured delivery with repository-scoped GitHub OIDC, RBAC, NetworkPolicies, and immutable commit-SHA promotion; validated drift self-healing and Git-revert recovery.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

GitHub

- Developed an EKS operations platform centralizing service ownership, deployment history, incident timelines, MTTR, and simplified error-budget views through FastAPI, React, PostgreSQL, and Alembic migrations.
- Deployed workloads behind ALB Ingress with Helm-managed probes and resource limits, structured JSON logs, request correlation IDs, Secrets Manager-backed credentials, and security-group-restricted RDS access.
- Validated HPA scale-out under k6 load as backend replicas increased from 2 to 6 pods, processing 2,035 requests with 99.5% success and 1.52s p95 latency.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

GitHub

- Built a serverless monitoring workflow where EventBridge invokes scheduled Lambda checks, DynamoDB stores current and historical status, SNS sends state-change alerts, and CloudFront serves a live React dashboard.
- Reduced current-status lookup candidates from 86,400 historical records to 10 latest-status rows for a 10-URL, 30-day reference workload; secured read-only API routes with throttling, restricted CORS, KMS, and unsafe-target blocking.
- Ran a controlled outage drill that detected failure in 1 second, displayed the DOWN state within 26 seconds, suppressed duplicate downtime alerts, and confirmed recovery notification.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India